Solve the following differential Equations

1. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$, given that $y(0) = 0$, $y'(0) = 15$.

Given eqn is

$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$

$(D^2 + 5D + 6)y = 0$

C.W

$f(D)y = Q(x)$

$f(D) = D^2 + 5D + 6$

$Q(x) = 0$

The A.E is $f(m) = 0$

$m^2 + 5m + 6 = 0$

$m^2 + 2m + 3m + 6 = 0$

$m(m+2) + 3(m+2) = 0$

$(m+2)(m+3) = 0$

$m = -2, m = -3$

$y_c = c_1e^{-2x} + c_2e^{-3x}$

$y_p = 0$

The G.S is

$y = y_c + y_p$

$y = c_1e^{-2x} + c_2e^{-3x} \rightarrow (1)$

Given $y(0) = 0 \Rightarrow x = 0, y = 0$

$0 = c_1e^{-2(0)} + c_2e^{-3(0)}$

$c_1 + c_2 = 0 \Rightarrow c_1 = -c_2$

$y' = -2c_1e^{-2x} - 3c_2e^{-3x}$

$y(0) = 15 \Rightarrow x = 0, y' = 15$

$15 = -2c_1e^{-2(0)} - 3c_2e^{-3(0)}$

$15 = -2c_1 - 3c_2$

$15 = -2c_1 - 3c_2$

$15 = -2(-c_2) - 3c_2$

$15 = -c_2 \Rightarrow \boxed{c_2 = -15}$

$\boxed{c_1 = 15}$

$\therefore$ The required soln is

$\boxed{y = 15e^{-2x} - 15e^{-3x}}$

2. $\frac{d^4x}{dt^4} + 4x = 0$

Given eqn is

$\frac{d^4x}{dt^4} + 4x = 0$.

$(D^4 + 4)x = 0$

C.W

$f(D)x = Q(t)$

$Q(t) = 0$

$f(D) = D^4 + 4$

The A.E is $f(m) = 0$

$m^4 + 4 = 0$

Now, add and substract $4m^2$

$(m^4 + 4m^2 + 4) - 4m^2 = 0$

$(m^2+2)^2 - (2m)^2 = 0$

$(m^2 + 2 + 2m)(m^2 + 2 - 2m) = 0$

For $m^2 + 2m + 2 = 0$

$m = \frac{-2 \pm \sqrt{4-8}}{2}$

$= \frac{-2 \pm \sqrt{-4}}{2}$

$m = \frac{-2 \pm 2i}{2} = -1 \pm i$

C.W $\alpha \pm \beta i \Rightarrow \alpha = -1, \beta = 1$

$y_c = e^{-x}[c_1\cos x + c_2\sin x]$

For $m^2 + 2 - 2m = 0$

$m = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$

$y_c = e^{x}[c_3\cos x + c_4\sin x]$

Here the function is in t

$y_p = 0$

$\therefore$ The G.S is

$y_c = e^{-t}[c_1\cos t + c_2\sin t] + e^{t}[c_3\cos t + c_4\sin t]$

③ $(D^2-6D+9)y = 6e^{3x}+7e^{-2x}-\log 2$

Given eqn is

$(D^2-6D+9)y = 6e^{3x}+7e^{-2x}-\log 2$

C.w $f(D)y = Q(x)$

$f(D) = D^2-6D+9$

$Q(x) = 6e^{3x}+7e^{-2x}-\log 2$

The A.E is $f(m)=0$

$m^2-6m+9=0$

$m^2-3m-3m+9=0$

$m(m-3)-3(m-3)=0$

$(m-3)(m-3)=0$

$m = 3,3$

$y_c = c_1e^{3x}+c_2e^{3x}$

$y_c = (c_1+c_2x)e^{3x}$

$y_p = \frac{Q(x)}{f(D)}$

$y_p = \frac{6e^{3x}+7e^{-2x}-\log 2}{D^2-6D+9}$

$y_p = \frac{6e^{3x}}{D^2-6D+9} + \frac{7e^{-2x}}{D^2-6D+9} - \frac{\log 2\cdot e^{0x}}{D^2-6D+9}$

$= \frac{6e^{3x}}{9-18+9} + \frac{7e^{-2x}}{4+12+9} - \frac{\log 2\cdot e^{0x}}{0-0+9}$

$= \frac{6e^{3x}}{0} + \frac{7e^{-2x}}{25} - \frac{\log 2\cdot e^{0x}}{9}$

$= \frac{6x\cdot e^{3x}}{2D-6} + \frac{7e^{-2x}}{25} - \frac{\log 2(1)}{9}$

$= \frac{6xe^{3x}}{0} + \frac{7e^{-2x}}{25} - \frac{\log 2}{9}$

$= \frac{6x^2e^{3x}}{2} + \frac{7e^{-2x}}{25} - \frac{\log 2}{9}$

$y_p = 3x^2e^{3x} + \frac{7e^{-2x}}{25} - \frac{\log 2}{9}$

The G.S is $y = y_c+y_p$

$$y = (c_1+c_2x)e^{3x} + 3x^2e^{3x} + \frac{7e^{-2x}}{25} - \frac{\log 2}{9}$$

4. $(D^2+5D-6)y = \sin 4x \sin x$

Given D.E is

$(D^2+5D-6)y = \sin 4x \sin x$

C.w $f(D)y = Q(x)$

$f(D) = D^2+5D-6$

$Q(x) = \sin 4x \sin x$

The A.E is $f(m)=0$

$m^2+5m-6=0$

$m^2+6m-m-6=0$

$m(m+6)-(m+6)=0$

$m = 1,-6$

$y_c = c_1e^{x}+c_2e^{-6x}$

$y_p = \frac{Q(x)}{f(D)} = \frac{2\sin 4x\sin x}{2(D^2+5D-6)}$

$y_p = \frac{\cos 3x - \cos 5x}{2(D^2+5D-6)}$

$y_p = \frac{1}{2}\left[\frac{\cos 3x}{D^2+5D-6} - \frac{\cos 5x}{D^2+5D-6}\right]$

Consider Term1:

$\frac{1}{2}\frac{1}{D^2+5D-6}\cos 3x$

sub $D^2 = -b^2 = -9$

$= \frac{1}{2}\frac{1}{-9+5D-6}\cos 3x$

$= \frac{1}{2}\frac{1}{(5D-15)}\cos 3x = \frac{\cdot 1}{10(D-3)}\cos 3x$

Rationalize by $D+3$

$= \frac{D+3}{10(D^2-9)}\cos 3x = \frac{(D+3)\cos 3x}{10(-18)}$

$= \frac{D\cos 3x + 3\cos 3x}{-180}$

$= \frac{-3\sin 3x + 3\cos 3x}{180}$

$= \frac{\sin 3x - \cos 3x}{6}$

consider term 2:

$$= \frac{-1}{2} \frac{1}{D^2+5D-6} \cos 5x$$

$$= \frac{-1}{2} \frac{1}{(-25+5D-6)} \cos 5x$$

$$= -\frac{1}{2(5D-31)} \cos 5x$$

Rationalise with $5D+31$

$$= \frac{-1}{2} \frac{5D+31}{(25D^2-961)} \cos 5x$$

$$= \frac{-1}{2} \frac{5D+31}{25(-25)-961} \cos 5x$$

$$= \frac{-1}{2} \frac{-25\sin 5x + 31\cos 5x}{(-1586)}$$

$$= \frac{31\cos 5x - 25\sin 5x}{3172}$$

$\therefore$ The G.S is

$$y = y_c + y_p$$

$$y = c_1 e^x + c_2 e^{-6x} + \frac{\sin 3x - \cos 3x}{60} + \frac{31\cos 5x - 25\sin 5x}{3172}$$

5. $(D^3+4D)y = 5+\sin 2x + 3^x$

Given eqn is

$$(D^3+4D)y = 5+\sin 2x + 3^x$$

c.w $f(D)y = Q(x)$

The A.E is $f(m) = 0$

$$m^3+4m = 0$$

$$m(m^2+4) = 0$$

$$m = 0 \quad m^2 = -4$$

$$m = \pm 2i$$

$$y_c = c_1 e^{0x} = c_1$$

$$y_c = e^{0x}[c_2 \cos 2x + c_3 \sin 2x]$$

$$y_c = c_1 + c_2\cos 2x + c_3 \sin 2x$$

$$y_p = \frac{5+\sin 2x + 3^x}{(D^3+4D)y}$$

$$= \frac{5}{D^3+4D} + \frac{\sin 2x}{D^3+4D} + \frac{e^{x\log 3}}{D^3+4D}$$

$$= \frac{5e^{0x}}{D^3+4D} + \frac{\sin 2x}{D(D^2+4)} + \frac{e^{x\log 3}}{D^3+4D}$$

$$= \frac{5e^{0x}}{0} + \frac{\sin 2x}{D(-4+4)} + \frac{e^{x\log 3}}{D^3+4D}$$

$$= \frac{5 \cdot x \cdot e^{0x}}{3D^2+4} + \frac{\sin 2x}{0} + \frac{e^{x\log 3}}{(\log 3)^3+4\log 3}$$

$$= \frac{5x}{4} + \frac{x\sin 2x}{3D^2+4} + \frac{e^{x\log 3}}{(\log 3)^3+4\log 3}$$

$$y_p = \frac{5x}{4} - \frac{x\sin 2x}{8} + \frac{e^{x\log 3}}{(\log 3)^3+4\log 3}$$

The G.S is

$y = y_c + y_p$

$$y = c_1 + c_2\cos 2x + c_3\cos 3x + \frac{5x}{4} - \frac{x\sin 2x}{8} + \frac{e^{x\log 3}}{(\log 3)^3+4\log 3}$$

6. $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2+2x+4$

Given D.E is

$\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2+2x+4$

$(D^2+D)y = x^2+2x+4$

C.W $f(D)y = Q(x)$

$f(D) = D^2+D$, $Q(x) = x^2+2x+4$

The A.E is $f(m)=0$

$m^2+m=0$

$m(m+1)=0$

$m=0 \quad m=-1$

$y_c = c_1e^{0x} + c_2e^{-x} \Rightarrow c_1 + c_2e^{-x}$

$$y_p = \frac{Q(x)}{f(D)} = \frac{x^2+2x+4}{D^2+D}$$

$$= \frac{x^2}{D^2+D} + \frac{2x}{D^2+D} + \frac{4e^{0x}}{D^2+D}$$

$$= \frac{x^2}{D(1+D)} + \frac{2x}{D(1+D)} + \frac{4e^{0x}}{0+0}$$

$$= \frac{x^2}{D}(1+D)^{-1} + \frac{2x}{D}(1+D)^{-1} + \frac{4xe^{0x}}{2D+1}$$

$$= \frac{x^2}{D}(1-D+D^2-\cdots) + \frac{2x}{D}(1-D+\cdots) + \frac{4xe^{0x}}{0+1}$$

$$= \frac{x^2-2x+2}{D} + \frac{2x-2}{D} + 4x$$

$$= \frac{x^3}{3} - \frac{2x^2}{2} + 2x + \frac{2x^2}{2} - 2x + 4x$$

$y_p = \frac{x^3}{3} + 4x$

∴ The G.S is

$y = y_c + y_p \Rightarrow y = c_1 + c_2e^{-x} + \frac{x^3}{3} + 4x$

7. $(D^2-4D+4)y = e^{2x}+8x^2$

Given DE is

$(D^2-4D+4)y = e^{2x}+8x^2$

c.w $f(D)y = Q(x)$

$f(D) = D^2-4D+4$

$Q(x) = e^{2x}+8x^2$

The AE is $f(m)=0$

$m^2-4m+4=0$

$m^2-2m-2m+4=0$

$m(m-2)-2(m-2)=0$

$(m-2)(m-2)=0$

$m=2,2$

$y_c = (c_1+c_2x)e^{2x}$

$y_p = \frac{f(D)}{Q(x)}$

$= \frac{e^{2x}+8x^2}{D^2-4D+4}$

$= \frac{e^{2x}}{D^2-4D+4} + \frac{8x^2}{D^2-4D+4}$

$= \frac{e^{2x}}{4-8+4} + \frac{8x^2}{4\left[1+\left(\frac{D^2-4D}{4}\right)\right]}$

$= \frac{e^{2x}}{0} + 2x^2\left[1+\left(\frac{D^2-4D}{4}\right)\right]^{-1}$

$= \frac{x\cdot e^{2x}}{2D-4} + 2x^2\left[1-\left(\frac{D^2-4D}{4}\right)+\left(\frac{D^2-4D}{4}\right)^2\right]$

$= \frac{x\cdot e^{2x}}{0} + 2x^2 - \left(4-16x\right) + \left(\frac{16}{4}\right)$

$= \frac{x^2e^{2x}}{2} + 2x^2 - 4 + 4x + 4$

$y_p = \frac{x^2e^{2x}}{2} + 2x^2+4x+3$

$\therefore$ The G.S is $y = y_c+y_p$

$y = (c_1+c_2x)e^{2x} + \frac{x^2e^{2x}}{2} + 2x^2+4x+3$

8. $(D^2-1)y = x\sin 3x + \cos x$

Given D.E is

$(D^2-1)y = x\sin 3x+\cos x$

c.w $f(D)y = Q(x)$

$f(D) = D^2-1 \quad Q(x) = x\sin 3x+\cos x$

The A.E is $f(m)=0$

$m^2-1=0$

$m^2=1$

$m=\pm 1$

$y_c = c_1e^x+c_2e^{-x}$

$y_p = \frac{x\sin 3x}{D^2-1} + \frac{\cos x}{D^2-1}$

$\rightarrow y_{p_1} \qquad \rightarrow y_{p_2}$

Consider

$y_{p_1} = \frac{x\sin 3x}{D^2-1}$

$= \frac{\text{I.P of } x\cdot e^{i3x}}{D^2-1}$

$= \left[\frac{\text{I.P of } e^{i3x}(x)}{(D+3i)^2-1}\right]$

$\therefore$ I.P of $\left[e^{i3x}\left(\frac{x}{D^2+6Di-10}\right)\right]$

$=$ I.P of $\left[\frac{e^{i3x}}{-10}\left(\frac{x}{1-\left(\frac{D^2+6Di}{10}\right)}\right)\right]$

$=$ I.P of $\left[\frac{e^{i3x}}{-10}x\left(1-\left(\frac{D^2+6Di}{10}\right)\right)^{-1}\right]$

$=$ I.P of $\left[\frac{e^{i3x}}{-10}\left(x\left[1+\frac{D^2+6Di}{10}+\cdots\right]\right)\right]$

$=$ I.P of $\left[\frac{e^{i3x}}{-10}\left(x+\frac{6i}{10}\right)\right]$

$=$ I.P of $\left[\frac{\cos 3x+i\sin 3x}{-10}\left(x+\frac{3i}{5}\right)\right]$

$= \frac{-1}{10}\left[\frac{3}{5}\cos 3x + x\sin 3x\right]$

$y_{p_1} = \frac{-1}{10}\left[\frac{3}{5}\cos 3x + x\sin 3x\right]$

$y_{p_2} = \frac{\cos^2 x}{D^2-1}$

Replace D^2 by -1^2

$y = y_c+y_p \Rightarrow y = c_1+c_2e^{\quad}+\frac{\quad}{3}+4x$

$$y_{p_2} = \frac{\cos^2 x}{-1} = \frac{-\cos^2 x}{2}$$

$$y_p = y_{p_1} + y_{p_2}$$

$$y_p = \frac{-1}{10}\left[\frac{3}{5}\cos 3x + x\sin 3x\right] - \frac{\cos^2 x}{2}$$

The G.S is $y = y_c + y_p$

$$y = c_1 e^{x} + c_2 e^{-x} - \frac{1}{10}\left[\frac{3}{5}\cos 3x + x\sin 3x\right] - \frac{\cos^2 x}{2}$$

$$y = c_1 e^{x} + c_2 e^{-x} - \frac{1}{10}\left[\frac{3}{5}\cos 3x + x\sin 3x\right] - \frac{\cos x}{2}$$

9. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2 e^{-3x}$ ...

Given DE is

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2 e^{-3x}$$

$$(D^2 + 2D - 3)y = x^2 e^{-3x}$$

C.W $f(D)y = Q(x)$

$f(D) = D^2 + 2D - 3$

$Q(x) = x^2 e^{-3x}$

The AE is $f(m) = 0$

$m^2 + 2m - 3 = 0$

$m^2 - m + 3m - 3 = 0$

$m(m-1) + 3(m-1) = 0$

$(m+3)(m-1) = 0$

$m = 1, -3$

$y_c = c_1 e^{x} + c_2 e^{-3x}$

$$y_p = \frac{Q(x)}{f(D)} = \frac{x^2 e^{-3x}}{(D^2 + 2D - 3)}$$

$$y_p = e^{-3x}\left[\frac{x^2}{(D-3)^2 + 2(D-3) - 3}\right]$$

$$y_p = e^{-3x}\left[\frac{x^2}{D^2 + 9 - 6D + 2D - 6 - 3}\right]$$

$$y_p = e^{-3x}\left[\frac{x^2}{-4D\left(1 - \frac{D}{4}\right)}\right]$$

$$= \frac{-e^{-3x}}{4}\left[\frac{1}{D}x^2\left(1 - \frac{D}{4}\right)^{-1}\right]$$

$$= \frac{-e^{-3x}}{4}\left[\frac{1}{D}x^2\left(1 + \frac{D}{4} + \frac{D^2}{16} + \cdots\right)\right]$$

$$= \frac{-e^{3x}}{4}\left[\frac{1}{D}\left(x^2 + \frac{2x}{4} + \frac{2}{16}\right)\right]$$

$$= \frac{-e^{3x}}{4}\left[\frac{x^3}{3} + \frac{x^2}{4} + \frac{x}{8}\right]$$

$$= \frac{-e^{3x}}{4}\left[\frac{8x^3 + 6x^2 + 3x}{24}\right]$$

$$y_p = \frac{-e^{3x}x(8x^2 + 6x + 3)}{96}$$

∴ The G.S is $y = y_c + y_p$

$$y = c_1 e^{x} + c_2 e^{-3x} - \frac{e^{3x}}{96}x(8x^2 + 6x + 3)$$

10. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x$

Given D.E is

$(D^2-2D+1)y = xe^x \sin x$

C.W $f(D)y = Q(x)$

The A.E is $f(m)=0$

$m^2-2m+1=0$

$m^2-m-m+1=0$

$m(m-1)-(m-1)=0$

$(m-1)(m-1)=0$

$m=1,1$

$y_c = (c_1+c_2x)e^x$

$y_p = \frac{xe^x \sin x}{D^2-2D+1}$

$= e^x\left[\frac{x\sin x}{(D+1)^2-2(D+1)+1}\right]$

$= e^x\left[\frac{x\sin x}{D^2+1+2D-2D-2+1}\right]$

$= e^x\left[\frac{x\sin x}{D^2}\right]$

$= e^x\left[\text{I.P of } \frac{x\cdot e^{ix}}{D^2}\right]$

$= e^x\left[\text{I.P of } \left(\frac{x\cdot e^{ix}}{D^2}\right)\right]$

$= e^x\left[\text{I.P of } e^{ix}\left(\frac{x}{(D+i)^2}\right)\right]$

$= e^x\left[\text{I.P of } -e^{ix}\left[\frac{x}{\left(1+\frac{D}{i}\right)^2}\right]\right]$

$= e^x\left[\text{I.P of } -e^{ix}\left(x\left(1+\frac{D}{i}\right)^{-2}\right)\right]$

$= e^x\left[\text{I.P of } -e^{ix}\left(x\left(1-\frac{2D}{i}+3\left(\frac{D}{i}\right)^2-\cdots\right)\right)\right]$

$= e^x\left[\text{I.P of } -e^{ix}\left(x-\frac{2}{i}+0\right)\right]$ $\quad -\frac{2}{i}\times\frac{i}{i} = -\frac{2i}{i^2} = \frac{2i}{1}$

$= e^x\left[\text{I.P of } -e^{ix}(x+2i)\right]$

$= e^x\left[\text{I.P of } -(\cos x+i\sin x)(x+2i)\right]$

$y_p = e^x\left[-2\cos x - x\sin x\right]$

$y_p = -e^x\left[x\sin x + 2\cos x\right]$

The G.S is $y = y_c + y_p$

$y = (c_1+c_2x)e^x - e^x\left[x\sin x + 2\cos x\right]$

11. $(D^2-4D+4)y = 8x^2e^{2x}\sin 2x$

Given D.E is

$(D^2-4D+4)y = 8x^2e^{2x}\sin 2x$

C.W $f(D)y = Q(x)$

The A.E is $f(m)=0$

$m^2-4m+4=0$

$m^2-2m-2m+4=0$

$m(m-2)-2(m-2)=0$

$(m-2)(m-2)=0$

$m=2,2$

$y_c = (c_1+c_2x)e^{2x}$

$y_p = \frac{8x^2e^{2x}\sin 2x}{D^2-4D+4}$

$= e^{2x}\left[\frac{8x^2\sin 2x}{(D+2)^2-4(D+2)+4}\right]$

$= 8e^{2x}\left[\frac{x^2\sin 2x}{D^2+4+4D-4D-8+4}\right]$

$= 8e^{2x}\left[\frac{x^2\sin 2x}{D^2}\right]$

$= 8e^{2x}\left[x^2\ \frac{\text{I.P of } e^{i2x}}{D^2}\right]$

$= 8e^{2x}\left[\text{I.P of}\left[e^{i2x}\left(\frac{x^2}{(D+2i)^2}\right)\right]\right]$

$= 8e^{2x}\left[\text{I.P of}\left[e^{i2x}\frac{x^2}{(2i)^2\left(1+\frac{D}{2i}\right)^2}\right]\right]$

$= 8e^{2x}\left[\text{I.P of } -4e^{i2x}\left[x^2\left(1+\frac{D}{2i}\right)^{-2}\right]\right]$

$= 8e^{2x}\left[\text{I.P of } -4e^{i2x}\left[x^2\left(1-2\left(\frac{D}{2i}\right)+3\left(\frac{D}{2i}\right)^2-\cdots\right)\right]\right]$

$= 8e^{2x}\left[\text{I.P of } -4e^{i2x}\left[x^2-\frac{2x}{i}+\frac{3(2)}{(4i)^2}\right]\right]$

$= 8e^{2x}\left[\text{I.P of } -4e^{i2x}\left[x^2-\frac{2x}{i}-\frac{6}{16}\right]\right]$

$= 8e^{2x}\left[\text{I.P of } -4(\cos 2x+i\sin 2x)\left(x^2-\frac{2x}{i}-\frac{3}{8}\right)\right]$

$= 8e^{2x}\left[8x\cos 2x-4\sin 2x\,x^2+\frac{3}{2}\sin 2x\right]$

$y_p = 4e^{2x}\left[16x\cos 2x-8\sin 2x\,x^2+3\sin 2x\right]$

∴ The G.S is

$y = y_c + y_p$

$y = (c_1+c_2x)e^{2x}+4e^{2x}\left[16x\cos 2x-8\sin 2x\,x^2+3\sin 2x\right]$

12. $y''-6y'+9y=\frac{e^{3x}}{x^2}$ by the method of variation of parameters.

Given D.E

$y''-6y'+9y=\frac{e^{3x}}{x^2}$

$(D^2-6D+9)y=\frac{e^{3x}}{x^2}$

C.W $f(D)y=Q(x)$

The A.E is $f(m)=0$

$m^2-6m+9=0$

$m^2-3m-3m+9=0$

$m(m-3)-3(m-3)=0$

$(m-3)(m-3)=0$

$m=3,3 \quad y_c=(c_1+c_2x)e^{3x}$

$y_c = c_1e^{3x} + c_2xe^{3x}$ c.w $y_c = c_1u(x) + c_2v(x)$

$u = e^{3x}$, $v = xe^{3x}$

$y_p = Ae^{3x} + Bxe^{3x}$

$u\frac{dv}{dx} - v\frac{du}{dx} = e^{3x}[3e^{3x}x + e^{3x}] - xe^{3x}[3e^{3x}]$

$= e^{6x} + 3e^{6x}x - 3xe^{6x}$

$= e^{6x}$

$A = -\int \frac{vR}{u\frac{dv}{dx} - v\frac{du}{dx}}dx = -\int \frac{x \cdot e^{3x} \cdot \frac{e^{3x}}{x^2}}{e^{6x}}$

$= -\int \frac{1}{x}dx$

$A = -\log x$

$B = \int \frac{uR}{u\frac{dv}{dx} - v\frac{du}{dx}}dx = \int \frac{e^{3x} \cdot \frac{e^{3x}}{x^2}}{e^{6x}}dx$

$= \int \frac{1}{x^2}dx$

$B = \frac{-1}{x}$

$y_p = -\log x \cdot e^{3x} - \frac{1}{x}xe^{3x}$

$y_p = -(e^{3x} + e^{3x}\log x)$

The G.S is $y = y_c + y_p$

$y = c_1e^{3x} + c_2xe^{3x} - e^{3x}\log x - e^{3x}$

13. $\frac{d^2y}{dx^2} + a^2y = \tan ax$ by the method of variation of parameters

Given D.E is

$\frac{d^2y}{dx^2} + a^2y = \tan ax$

$(D^2 + a^2)y = \tan ax$

c.w $f(D)y = \phi(x)$

The A.E is $f(m) = 0$

$m^2 + a^2 = 0$

$m^2 = -a^2$

$m = \pm ai$

$y_c = c_1\cos ax + c_2\sin ax$

c.w $y_c = c_1u(x) + c_2v(x)$

$u(x) = \cos ax$ $v(x) = \sin ax$

$y_p = A\cos ax + B\sin bx$

$u\frac{dv}{dx} - v\frac{du}{dx} = \cos ax(a\cos ax) + \sin ax(a\sin ax) = a(\cos^2 ax + \sin^2 ax) = a$

$A = \frac{-1}{a}\int \sin ax \tan ax\, dx$

$= \frac{-1}{a}\int \frac{\sin^2 ax}{\cos ax}\, dx$

$= \frac{-1}{a}\int \frac{1-\cos^2 ax}{\cos ax}\, dx$

$= \frac{-1}{a}\int (\sec ax - \cos ax)\, dx$

$= \frac{-1}{a}\left[\frac{1}{a}\log(\sec ax + \tan ax) - \frac{1}{a}\sin ax\right]$

$= \frac{-1}{a^2}\left[\log(\sec ax + \tan ax) - \sin ax\right]$

$A = \frac{-1}{a^2}\left[\log(\sec ax + \tan ax) + \sin ax\right]$

$B = \frac{1}{a}\int \cos ax \tan ax\, dx$

$= \frac{1}{a}\int \sin ax\, dx$

$B = \frac{-\cos ax}{a}$

$y_p = \frac{-1}{a^2}(\log(\sec ax + \tan ax) - \sin ax)\cos ax - \frac{\cos ax}{a}\sin ax$

The GS is $y = y_c + y_p$

$y = C_1 \cos ax + C_2 \sin ax - \frac{1}{a^2}\left[\log(\sec ax + \tan ax) - \sin ax\right] - \cos ax - \frac{\cos ax}{a}\sin ax$